

RANJITH LK

M.Sc Decision and Computing Sciences

☎ 9489766467 ✉ ranjithsabapathy2004@gmail.com  [Ranjith LK](#)  [Ranjith352](#)

PROFILE

As a passionate and motivated student, I am eager to apply my academic knowledge and technical skills in a dynamic and challenging environment. I enjoy solving problems, collaborating with teams, and exploring innovative solutions to real-world challenges. With strong adaptability and a continuous learning mindset, I am committed to contributing effectively while growing both personally and professionally. Additionally, my adaptability and eagerness to continuously learn equip me to tackle unfamiliar situations with confidence.

PROJECTS

Healthcare Prediction System (May 2024): Built a healthcare prediction platform using Predictive Analytics with Streamlit and MySQL integration. Added interactive visualizations and personalized dashboards to provide health insights and recommendations.

AI-Based Pothole Detection System (Apr 2025): Developed an AI-based pothole detection system using OpenCV, CNN (TensorFlow/Keras), and YOLOv8 for accurate pothole classification and localization. Built a React and FastAPI web application with image upload, webcam detection, LiDAR-style visualization, and automated PDF reporting for intelligent road monitoring.

AI-Powered Complaint Management System (April 2026): Built a full-stack AI-powered complaint management system using React, FastAPI, PostgreSQL, NLP, Hugging Face, and pgvector for intelligent complaint classification, sentiment analysis, duplicate detection, and team routing. Integrated Gmail API, Ollama/Groq, RAG, AI-assisted responses, SLA tracking, and analytics to streamline complaint resolution and customer support.

The Lenny Growth Assistant – AI-Powered Conversational Growth Assistant (June 2026): Developed a full-stack AI-powered conversational assistant using FastAPI, React, PostgreSQL, and RAG over Lenny's Podcast transcripts. Integrated OpenAI, Groq, and Ollama with runtime provider switching, semantic vector search (pgvector), and interactive Markdown/HTML artifact generation.

WORK EXPERIENCE

Full Stack Developer Intern, QR Solutions, Trichy (Jun 2025 – Nov 2025):

Worked on developing and maintaining full-stack web applications by integrating frontend interfaces with backend APIs and databases. Designed responsive user interfaces and implemented backend logic to support dynamic features and efficient data management. Collaborated with the development team on debugging, testing, deployment, and performance optimization of web applications.

EDUCATION

Higher Secondary Education (2021 – 2022): Scored 83% in 12th Grade

M. Sc Decision and Computing Sciences (2022 – 2027): CGPA: 7.9 as of June 2026

ADDITIONAL INFORMATION

Technical Skills

Python, R, Full Stack Development, Mobile and Cloud Application Development, DevOps, Data Analysis, NLP

Database Management

MySQL, MongoDB, PostgreSQL, Firebase

Certifications:

- Presented at Voyage Hackathon in Vellore Institute of Technology (2024)
- Workshop on “Large Language Models and Their Applications” by Microsoft Virtual internship in Artificial Intelligence at CodSoft(2024)
- 2nd Prize in Melinia Hackathon in Bid Battle (2024)
- Byte Sized Brainstorm Hackathon (2023)
- Hackrush Hackathon By WEC (2025)